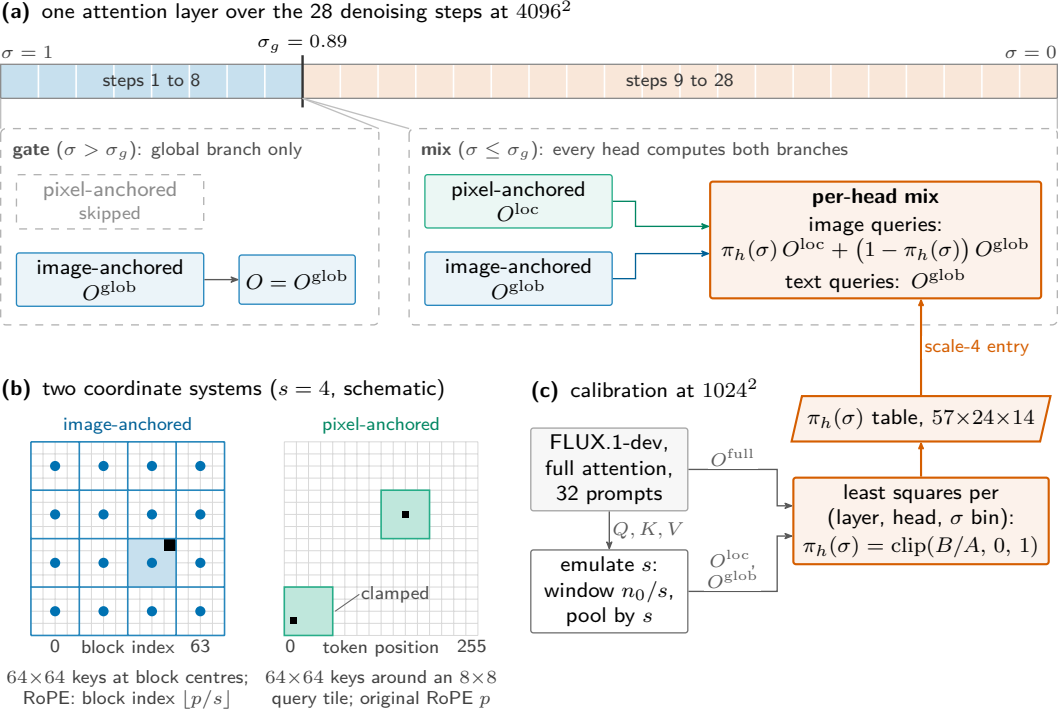
(a) one attention layer over the 28 denoising steps at 4096^2
$\sigma = 1$
$\sigma_g = 0.89$
$\sigma = 0$
steps 1 to 8
steps 9 to 28
gate ($\sigma > \sigma_g$): global branch only
pixel-anchored
skipped
image-anchored
O^{glob}
$O = O^{\text{glob}}$
mix ($\sigma \le \sigma_g$): every head computes both branches
pixel-anchored
O^{loc}
image-anchored
O^{glob}
per-head mix
image queries:
$\pi_h(\sigma)\, O^{\text{loc}} + (1 - \pi_h(\sigma))\, O^{\text{glob}}$
text queries: O^{glob}
scale-4 entry
(b) two coordinate systems ($s = 4$, schematic)
image-anchored
pixel-anchored
0
block index
63
clamped
0
token position
255
64×64 keys at block centres;
RoPE: block index $\lfloor p/s \rfloor$
64×64 keys around an 8×8
query tile; original RoPE p
(c) calibration at 1024^2
FLUX.1-dev,
full attention,
32 prompts
O^{full}
Q, K, V
emulate s:
window n_0/s,
pool by s
O^{loc},
O^{glob}
least squares per
(layer, head, σ bin):
$\pi_h(\sigma) = \text{clip}(B/A,\, 0,\, 1)$
$\pi_h(\sigma)$ table, $57\times24\times14$